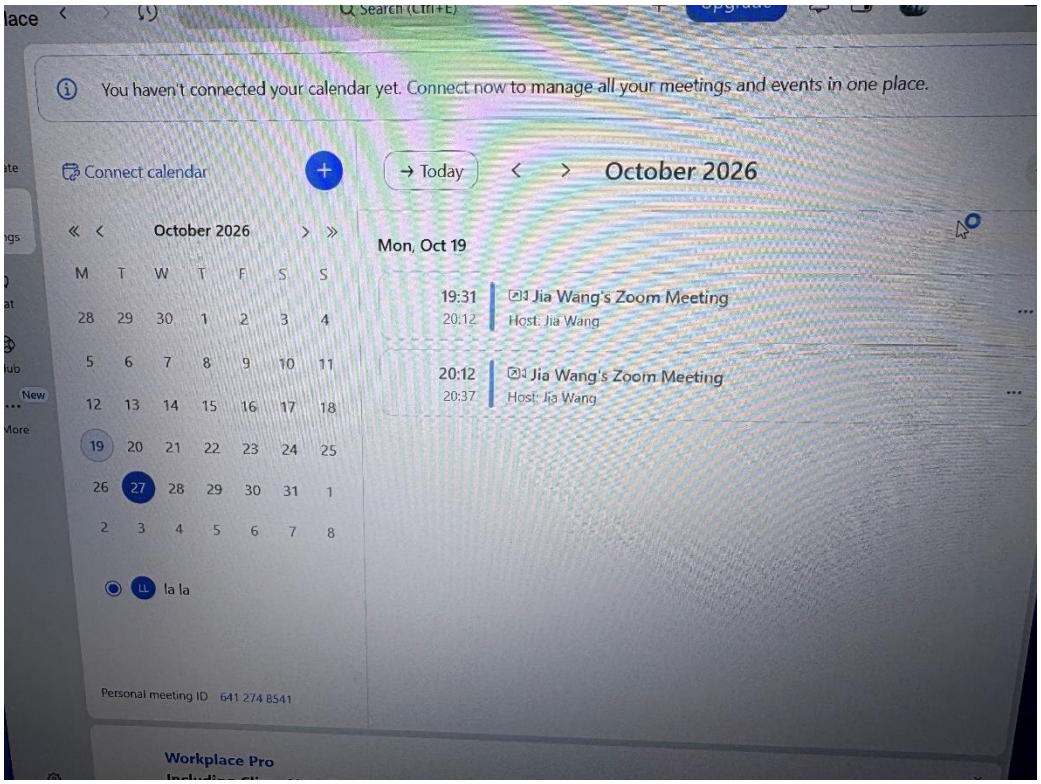
I came up with the **basic concept and initial idea** for this project. But I was thinking an interesting board/ card game for students living in a share house to help them split the shared expense at beginning, but we moved from physical card game to a mobile website after team discussion.

I organized team discussion sessions and zoom meetings, I briefed the team and tracked our progress.



The **conference poster was designed and created by me**.

DECO3500 TEAM 17 SHAREHOUSE EXPENSES

The Weekly Split

A shared ritual for agreeing on household expenses

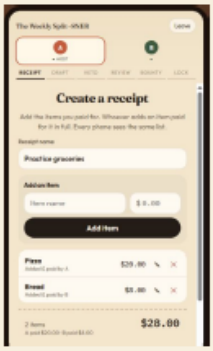
In sharehouses, questioning a small grocery cost can feel more awkward than paying it. Our opportunity is to make that disagreement easier to raise.

01 INTERVIEWS SHAPED THE OPPORTUNITY

17 problem interviews: people described accepting small unfair costs to avoid tension. We designed item-level choices and review before repayment, so housemates can express their use without opening with an accusation. [1]

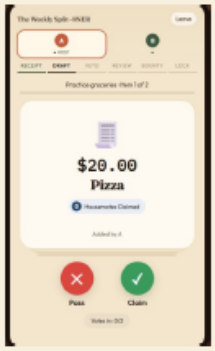
Supporting survey: 29 of 35 respondents at least occasionally paid equally despite consuming much less. [2]

02 2–8 HOUSEMATES, ONE LIVE TABLE



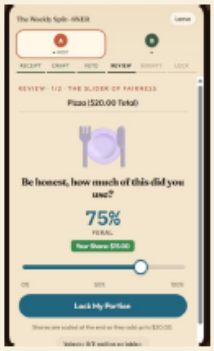
1 Create a receipt

Add purchases you paid for. Everyone contributes items and prices to one shared receipt.



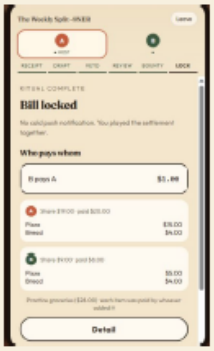
2 Claim / Pass

Choose which items you use. Claimants share each item's cost.



3 Review together

Raise a concern, discuss it, then slide your estimated use from 0–100%.



4 Explain

See who pays whom. Open Detail to trace paid amounts and item shares.

03 WHY SOCIAL & MOBILE COMPUTING?

Social: the 3C model

Cooperation: build one receipt. Communication: claims and review express use and concerns. Coordination: live votes and readiness pace the group. Awareness: see who has responded. [6, 8]

Mobile: the household context

A Sunday kitchen-table ritual fits shared shopping, household relationships and weekly routines. Separate phones support individual input during a same-time session. [7]

04 USER TESTS CHANGED THE DESIGN

Users rejected orphan penalties. We removed them.

Randomly assigning an unclaimed expense felt unfair to the person selected. We removed the orphan-item penalty game after user tests. Unclaimed expenses now split evenly; agreement still matters. [3, 4, 5]

Keep item-level choices

In one four-person household test, all four completed the main flow without hints; three traced an item line behind their amount. [3]

Visibility can create pressure

Earlier tests exposed veto inference. Lettered seats and hidden responses now support discretion; social privacy still needs evaluation. [3, 5, 6]

NEXT: household groups in context; observe participation, pressure and repeated use

[1] Interviews [2] Survey [3] Team user tests [4] Team decision [5] Prototype
[6] Fuks et al., 3C (Week 2) [7] Context, Weeks 1 & 6 [8] Erickson, IxDF

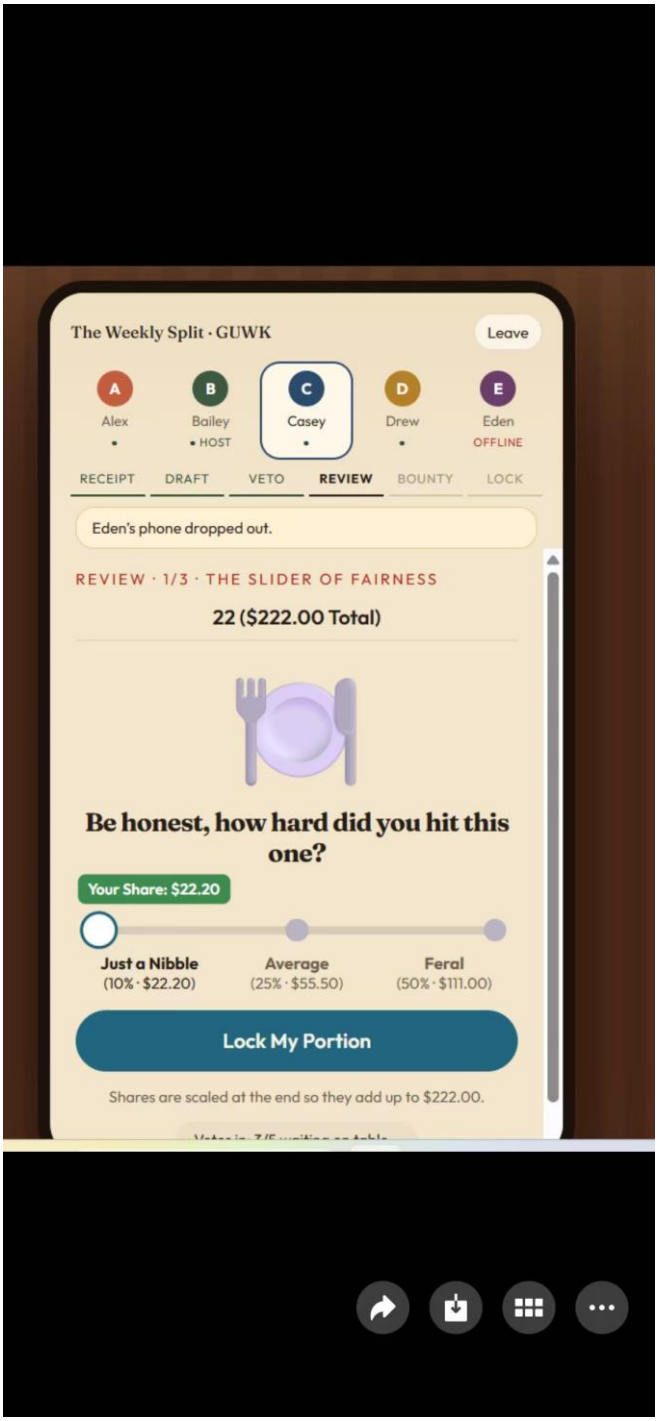
Join a table. Try your own split.

malonewayne.github.io/Deco3500-Team-17/

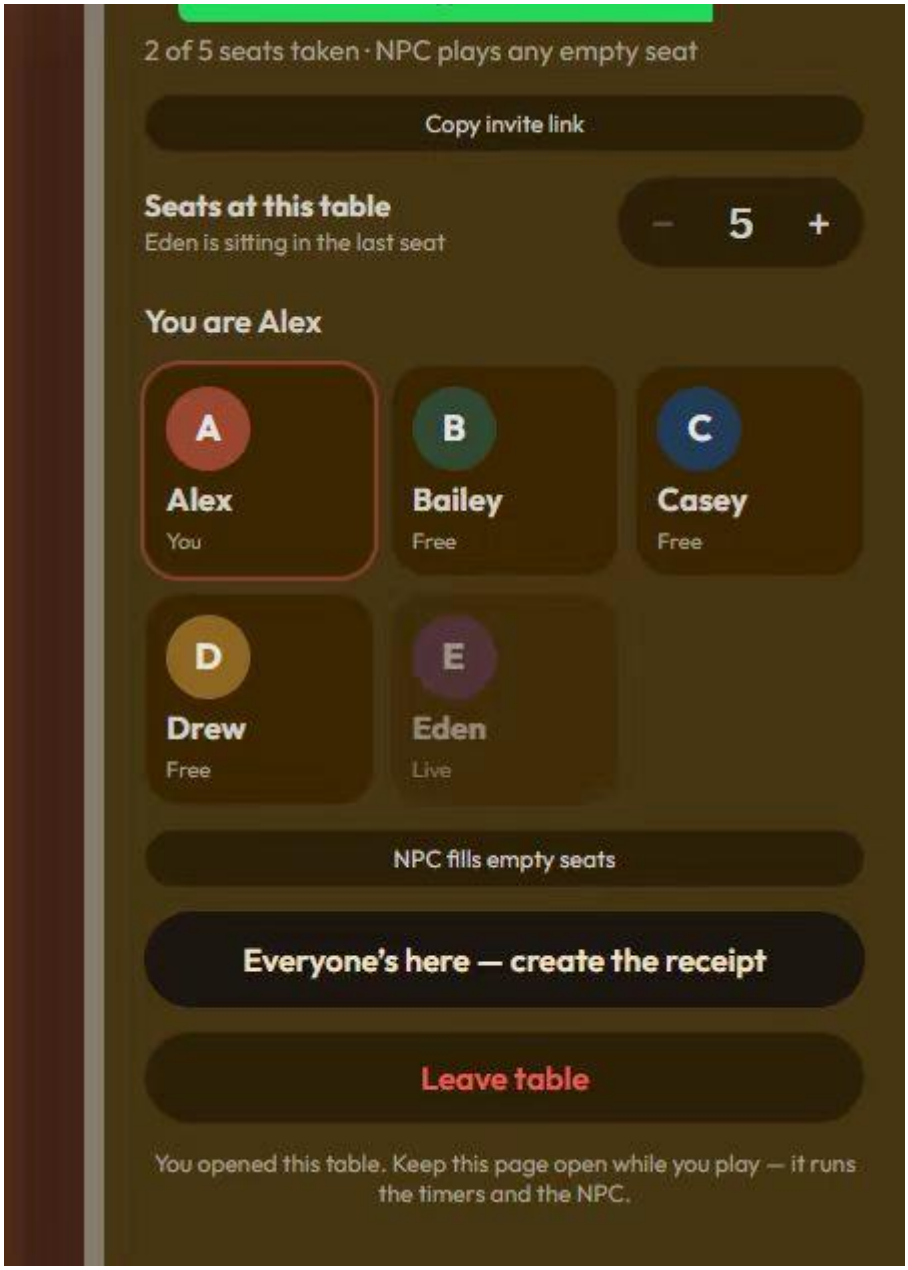
Team research: github.com/MaloneWayne/Deco3500-Team-17

The next part is about the **prototype improvements and design decisions**:

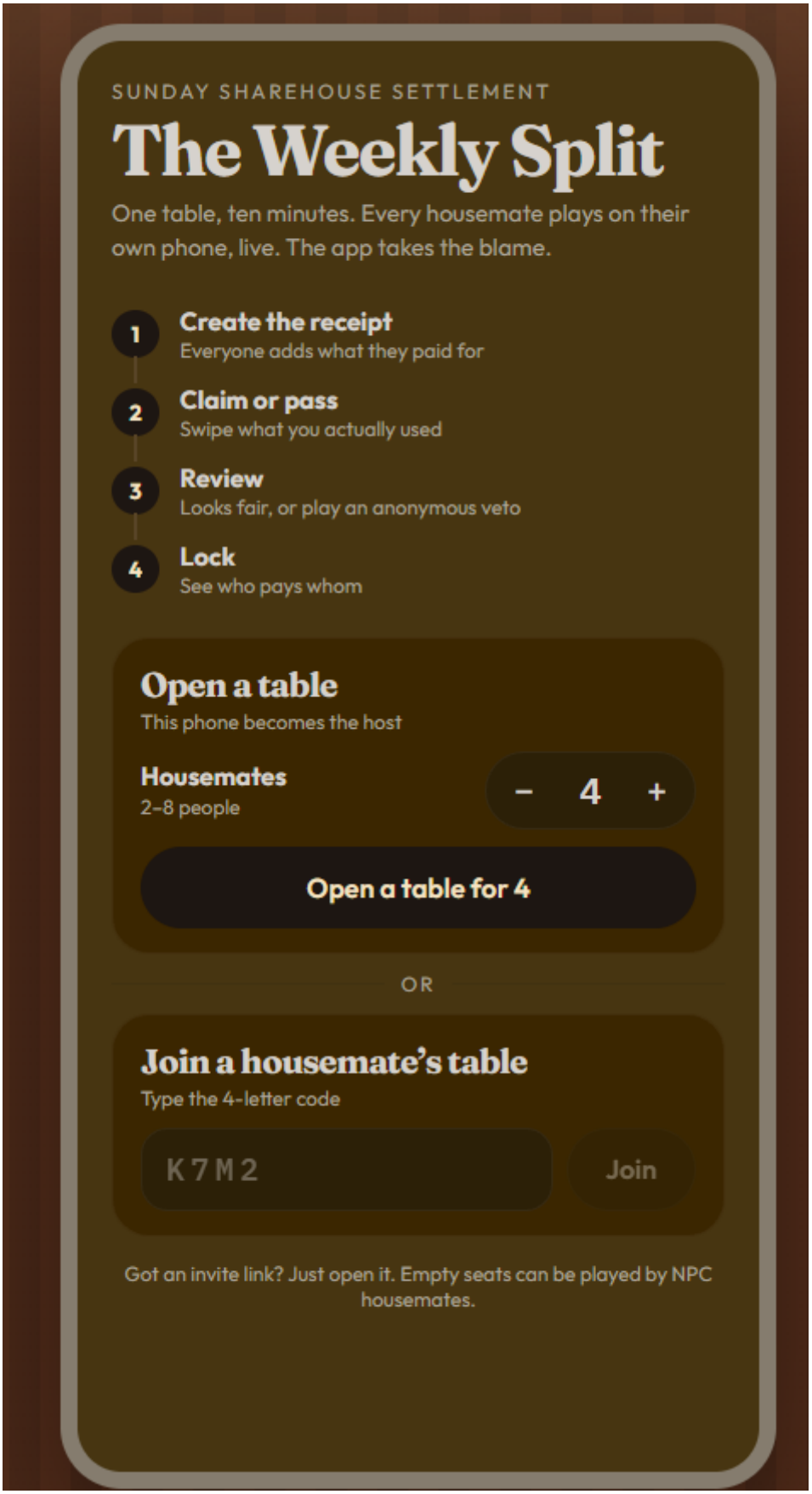
I suggested allowing users to select a percentage from 0% to 100% when estimating their consumption.



instead of using 'Alex' 'Bailey', we should simply use ABCDE and customizd name to represent users.

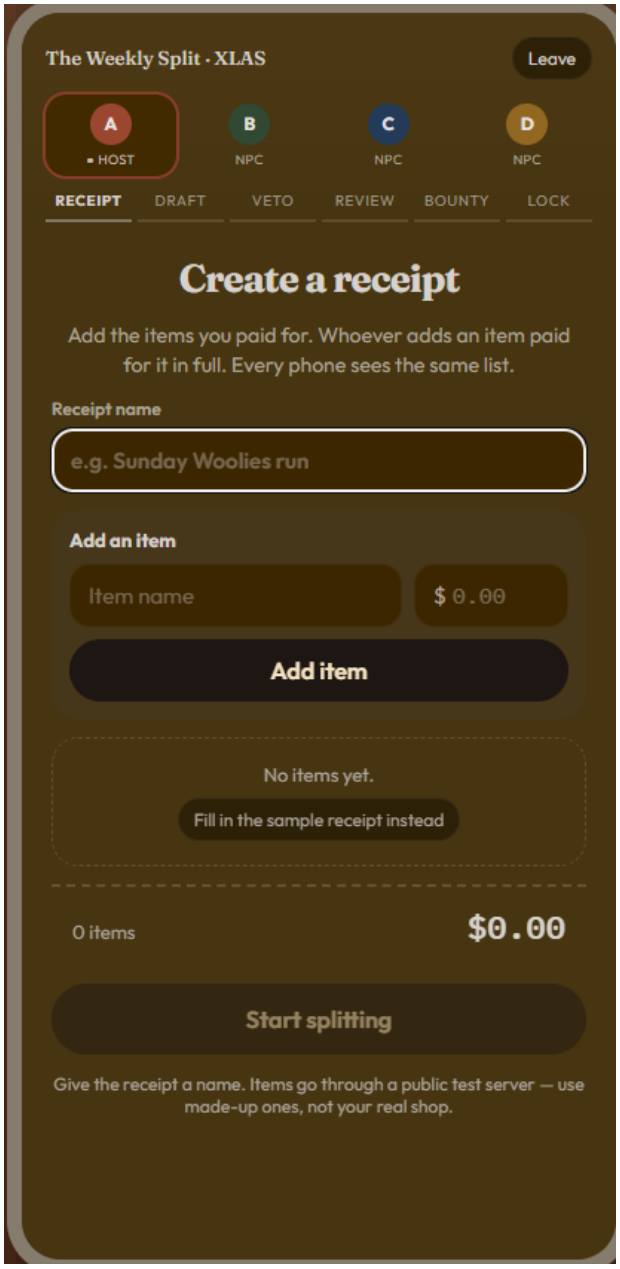


Added a custom participant count feature; previously, only 4 or 5 people were allowed.



adding “Create a receipt” before the Claim/Pass stage:

1. Enter a receipt name and select who paid.
2. Add each item’s name and price.
3. Edit or remove items and check the calculated total.
4. Tap “Start splitting” to continue.



Removed redundant content from the homepage.